

Computer Engineering undergraduate at the National University of Singapore with hands-on experience in embedded systems, hardware validation, robotics, FPGA design, AI, and software engineering.

EDUCATION

National University of Singapore

B.Eng. Computer Engineering · Second Major, Innovation and Design Programme

Aug 2024—Jul 2028

GPA: 4.13 / 5.0

Relevant coursework:

Data Structures & Algorithms; Software Engineering & OOP; Digital Design; Computer Organization; Signals & Systems; Introduction to Machine Learning.

Current:

Real-Time Operating Systems; Computer Architecture; Human-Robot Interaction; Guidance, Navigation & Control.

TECHNICAL SKILLS

Programming & RTL	C, C++, Python, Java, JavaScript, GDScript, Verilog, ARM Thumb Assembly	Embedded & hardware	STM32, ARM Cortex-M4, FreeRTOS, bare-metal firmware, CAN/FDCAN, UART, I ² C, GPIO, timers/PWM, NFC/NDEF, HIL testing, PCB design, FPGA
Robotics & AI	ROS 2, MuJoCo, Nav2, SLAM Toolbox, TurtleBot3, inverse kinematics, autonomous navigation, PyTorch, Hugging Face, OpenCV, vision-language models	Software & tools	Node.js, Express, MongoDB, JUnit 5, Git, Docker, Linux, STM32CubeIDE, Vivado, Vitis, KiCad, LTspice, Godot 4, PlantUML

EXPERIENCE

Embedded Systems Intern

LionsBot International

May—Jul 2026

- Architected and led development of a hardware-in-the-loop ECU test platform using STM32U585, C, FreeRTOS, FDCAN, and Python/pywebview to emulate sensors, actuators, loads, current feedback, and pulse signals.
- Automated approximately 20 end-to-end validation cases, reducing execution from approximately 10 minutes to under 1 minute; designed the architecture to expand toward approximately 200 planned cases.
- Independently developed a Python/Tkinter R5 RFS QC application for UART execution, response parsing, pass/fail evaluation, operator cross-verification, and traceable HTML reports.
- Reduced RFS validation from approximately 15–20 minutes to approximately 2–5 minutes per unit across approximately 12–15 hardware components; deployed the workflow to the manufacturer in China.
- Designed, fabricated, and validated a two-layer KiCad trolley-interface PCB, including component selection, layout, bring-up, rapid prototyping, and mechanical fit checks.
- Analyzed and simulated robot PDB and ECU circuitry in LTspice, including fans, buffers, opto-isolators, I²C level shifters, and interface circuits.

Teaching Assistant

National University of Singapore · CG2111A and CDE2310

Jan—Apr 2026

- Guided approximately 20 students through bare-metal Arduino labs, C, register-level peripheral configuration, I/O, and debugging; assessed lab and project work.
- Mentored 12 student teams using ROS 2 on Ubuntu, including packages, workspace setup, Nav2 usage, build and launch troubleshooting, presentations, and mission runs.

Vice President, External

NUS Bitscraps

May 2026—Present

- Oversee Publicity and Events for a sustainability-focused electronics repair club; coordinate event execution, promotion, registrations, member communication, website content, and external outreach.

SELECTED PROJECTS

Language-Guided Humanoid Manipulation System Sep 2026—Present

MuJoCo · Vision-Language AI · Robotics

- Built a Unitree G1 MuJoCo environment and integrated a local Qwen2.5-VL-7B model using PyTorch, Hugging Face Transformers, 4-bit NF4 quantization, Docker, and persistent model caching on an 8 GB NVIDIA GPU.
- Combined vision-language grounding with OpenCV refinement, improving grounding accuracy from 54.2% to 91.7% across 24 trials; achieved 24/24 validated language-to-action planning outcomes.
- Implemented inverse-kinematics motion, contact-aware grasping, object tracking, controlled release, recovery, and visual verification; completed 12/12 randomized manipulation trials with 3 cm mean placement error.
- Built a live dashboard and reproducible evaluation framework for 20 randomized scenarios with JSONL evidence, system metrics, and 33 automated tests.

Vision-Language Model Adversarial Robustness Benchmark Sep 2026—Present

Qwen2.5-VL · PyTorch · OpenCV

- Built a deterministic 50-item A-OKVQA evaluation pipeline with pinned local inference, stable IDs, raw responses, hashes, manifests, environment metadata, and report-ready metrics.
- Designed a visual prompt-injection attack that reduced accuracy from 90% to 76%, then an image-only defense that recovered all 8 successful attacks with 0 false detections across 50 clean images.

Wearable Fall Detection Device Feb—Mar 2026

STM32L455 · C · ARM Thumb · NFC

- Developed a five-state fall detector using accelerometer and gyroscope data, jitter rejection, timing thresholds, and a 10-second long-lie check.
- Wrote a four-sample X/Y/Z moving-average filter in ARM Thumb assembly and exported fall count, uptime, and timestamps as NDEF URI data for a phone dashboard.

K-Map Simplifier and Visualiser Oct—Nov 2025

Basys 3 FPGA · Verilog · OLED

- Built switch-based truth-table input, SOP/POS modes, canonical-expression generation, reusable button conditioning, and mode-aware top-level integration.
- Developed a 96×64 OLED pixel pipeline with a 5×7 character renderer, Gray-code K-Map, color themes, and dynamic cell highlighting.

Autonomous Search-and-Rescue Robot Jan—Apr 2025

TurtleBot3 · ROS 2 · Nav2 · SLAM

- Integrated SLAM Toolbox, Nav2, LiDAR, an AMG8833 thermal array, and GPIO/PWM actuation on a Raspberry Pi 4.
- Extended frontier exploration to interrupt for heat detections, avoid handled locations, and resume navigation; tuned the physical robot and built isolated sensor/actuator test tools.

UniQuest May—Aug 2025

Godot 4 · GDScript · Android · Web

- Co-developed a two-person 2D platformer with reusable character state machines, combat, weapons, progression, touch controls, customization shaders, and cross-platform exports.
- Built JSON persistence with migration support and GUT tests; created original pixel art, background music, and sound.

ADDITIONAL WORK & CREDENTIALS

CuddleCare — Java command architecture, parsing, Streams, defensive persistence, structured logging, JUnit 5, and PlantUML.

GARUDA — Node.js/Discord.js bot with modular commands, MongoDB configuration, persistent giveaways, moderation, and community tools.

JavaCS — Node.js/Express archive for 30 Java solutions with JSON-backed content, search, filters, and dynamic routes.

Certifications: Google Data Analytics; AI Singapore AI4I Foundations in AI; Orbital Apollo 11 (Advanced); PSA Code Sprint.